

CONSTRAINT MOTION IN PULLEY SYSTEMS

JEE Advanced Preparation — Physics

Topic 02 | Newton's Laws, Atwood Machine, Movable Pulleys, Constraint Equations

Concept	Key Relation	Application
Inextensible string	$d(\text{length})/dt = 0$	Atwood machine
Fixed pulley	$a_1 = a_2$ (magnitude)	Changes force direction
Movable pulley	$a_{\text{mass}} = 2a_{\text{pulley}}$	$MA = 2$
n movable pulleys	$MA = 2^n$	Block and tackle
Constraint equation	Sum of velocities along string = 0	Complex systems
Atwood tension	$T = 2m_1m_2g/(m_1+m_2)$	Always between m_1g and m_2g

Constraint motion problems require identifying the physical restriction on motion and expressing it mathematically. The constraint equation links the motions of all bodies in the system and is derived by differentiating the total string length equation.

KEY FORMULAS REFERENCE

String constraint: $L_{\text{total}} = \text{const} \Rightarrow \text{sum}(\text{velocity components along string}) = 0$

Atwood machine: $a = (m_2 - m_1)g / (m_1 + m_2)$ [$m_2 > m_1$, a is acceleration of m_2 downward]

Atwood tension: $T = 2m_1m_2g / (m_1 + m_2)$ [always: $m_1g < T < m_2g$]

Movable pulley: $a_{\text{load}} = 2a_{\text{pulley}}$; $T_{\text{string}} = W_{\text{load}} / 2$ (for equilibrium)

Inclined constraint: $a_{\text{block along incline}} = a_{\text{hanging mass}}$ (inextensible string)

Massive pulley (disk): $a = (m_2 - m_1)g / (m_1 + m_2 + M_{\text{pulley}}/2)$

n supporting rope segments: $T = W/n$; $MA = n$ (mechanical advantage)

Component method: $v \cos(\theta) = \text{rate of change of string length for inclined strings}$

SECTION 1: FUNDAMENTALS OF CONSTRAINT EQUATIONS

A constraint is any physical restriction that limits the motion of a body. For strings and pulleys, the primary constraint is the inextensibility of the string — its total length does not change.

Method of Constraint Equations (Step-by-Step):

Step 1: Define coordinates. Assign a position variable to each moving body. Choose a reference (usually ceiling or fixed wall) and measure downward (or along motion direction).

Step 2: Write the total string length equation.
For each segment of string, express its length in terms of the position variables.
Set the sum equal to a constant: $L_1 + L_2 + \dots + \text{constant_wraps} = L_{\text{total}}$.

Step 3: Differentiate once with respect to time $\Rightarrow$ velocity constraint.
 $v_1 + v_2 + \dots = 0$ (sign depends on sign convention chosen).

Step 4: Differentiate again $\Rightarrow$ acceleration constraint.
 $a_1 + a_2 + \dots = 0$

Sign Convention:

If you define positive as downward for all masses:

Downward motion increases position variable $\Rightarrow$ positive velocity contribution.

Upward motion decreases position variable $\Rightarrow$ negative contribution.

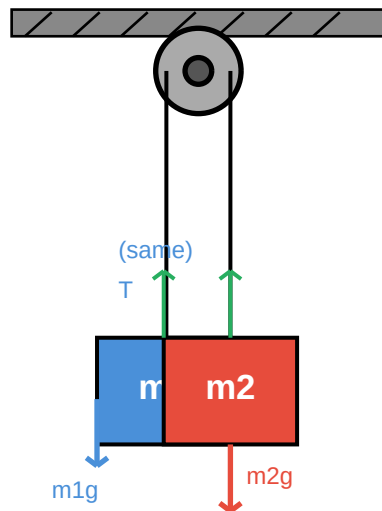
The constraint equation automatically handles signs if you are consistent.

Pulley wrap: String wrapped around a pulley contributes a fixed constant to length. Only the hanging segments vary with time.

KEY INSIGHT: The constraint equation is purely KINEMATIC. It is independent of forces, masses, or Newton's laws.

It only reflects the geometry of the string. Forces are handled separately via Newton's second law.

SECTION 2: ATWOOD MACHINE — COMPLETE ANALYSIS



Constraint:

$$x_1 + x_2 = L = \text{const}$$

$$a_1 = -a_2 \text{ (equal magnitude)}$$

$$a = (m_2 - m_1)g / (m_1 + m_2)$$

$$T = 2m_1m_2g / (m_1 + m_2)$$

Fig 1: Atwood Machine — two masses over fixed pulley

Setup: Two masses m_1 and m_2 ($m_2 > m_1$) connected by massless inextensible string over a fixed frictionless massless pulley.

Step 1 — Constraint:

Let x_1 = distance of m_1 below ceiling, x_2 = distance of m_2 below ceiling.

$$x_1 + x_2 + \text{constant (wraps)} = L_{\text{string}}$$

$$\text{Differentiating: } v_1 + v_2 = 0 \Rightarrow v_1 = -v_2$$

So if m_2 moves down ($v_2 > 0$), m_1 moves up ($v_1 = -v_2 < 0$).

$a_1 = -a_2$. Let magnitude = a (both have same magnitude).

Step 2 — Newton's Laws:

For m_2 (take downward as positive): $m_2g - T = m_2a$... (i)

For m_1 (downward positive, m_1 accelerates upward): $T - m_1g = m_1a$... (ii)

Note: m_1 accelerates upward = negative downward = $-a$, so from downward+ perspective: $T - m_1g = m_1a$

Step 3 — Solve:

$$\text{Add (i) and (ii): } (m_2 - m_1)g = (m_1 + m_2)a$$

$$a = (m_2 - m_1)g / (m_1 + m_2)$$

$$\text{Substitute in (ii): } T = m_1(g + a) = m_1g + m_1(m_2 - m_1)g / (m_1 + m_2) = 2m_1m_2g / (m_1 + m_2)$$

Checks and insights:

If $m_1 = m_2 = m$: $a = 0$, $T = mg$. System is in equilibrium; tension equals weight.

If $m_2 \gg m_1$: $a \rightarrow g$ (free fall), $T \rightarrow 2m_1g$ (small tension).

T is always between m_1g and m_2g .

Heavier mass provides driving force; lighter mass provides resistance.

Both masses have same $|a|$ and same T by constraint.

SECTION 3: MOVABLE PULLEY SYSTEMS

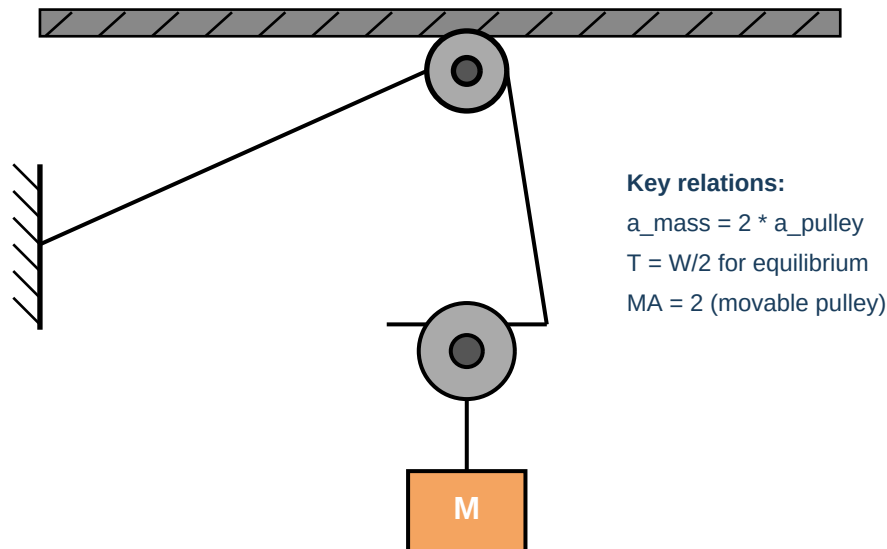


Fig 2: Movable Pulley — mechanical advantage 2

Key Principle: When a movable pulley descends by x , both string segments over it shorten by x each.

Therefore, the free string end must extend by $2x$ to compensate.

=> If pulley moves down at velocity v_p , free string end moves at $2*v_p$.

=> $a_{\text{free_end}} = 2 * a_{\text{pulley}}$.

Force analysis for massless movable pulley:

String tension T is uniform (massless string, frictionless pulley).

Two string segments pull the pulley upward: net upward string force = $2T$.

For the load W attached to pulley: $W = 2T$ (equilibrium) => $T = W/2$.

Mechanical advantage = $W/T = 2$. Half the force needed, but twice the distance.

General n-segment rule (compound pulley system):

If n string segments support the movable pulley/load: $T = W/n$; $MA = n$.

But: string end must move n times the load displacement.

Efficiency = (useful work out) / (work in) = 1 for ideal (frictionless) pulleys.

Dynamic analysis (mass M on movable pulley, string over fixed pulley, free end pulled with force F):

Let a_M = acceleration of M , a_p = acceleration of fixed-pulley end of string.

Constraint: $a_M = 2 * a_p$ (if string end over fixed pulley moves at a_p).

For M : $M*g - 2T = M*a_M$

For string end: $F - T = m_{\text{string}} * a_p$ (massless: $F = T$, so if F varies...)

Usually: free end is pulled with $T = F$. Then: $M*g - 2F = M*a_M$ => $a_M = g - 2F/M$.

SECTION 4: MULTIPLE PULLEY SYSTEMS — CONSTRAINT ANALYSIS

System with one fixed pulley (P1) and one movable pulley (P2):

String 1 passes over P1 (fixed), connecting mass m_1 to axle of P2.

String 2 passes over P2 (movable), connecting mass m_2 to mass m_3 .

Let: y_1 = position of m_1 (down+), y_{P2} = position of P2 (down+), y_2 = pos of m_2 , y_3 = pos of m_3 .

String 1 constraint: $y_1 + y_{P2} = C_1 \Rightarrow v_1 = -v_{P2} \Rightarrow a_1 = -a_{P2}$

String 2 constraint: $(y_2 - y_{P2}) + (y_3 - y_{P2}) = C_2 \Rightarrow v_2 + v_3 - 2v_{P2} = 0$

$\Rightarrow v_2 + v_3 = 2v_{P2} = -2v_1$

$\Rightarrow a_2 + a_3 = -2a_1$

Three equations needed:

(1) Constraint: $a_2 + a_3 + 2a_1 = 0$

(2) Newton's law for m_1 : $m_1g - T_1 = m_1a_1$ (T_1 = tension in string 1)

(3) For P2 (massless): $T_1 = 2T_2$ (string 2 tension = T_2 , two segments on P2)

(4) For m_2 : $m_2g - T_2 = m_2a_2$

(5) For m_3 : $m_3g - T_2 = m_3a_3$

Five equations, five unknowns (a_1, a_2, a_3, T_1, T_2). Solve simultaneously.

Component method for inclined strings:

If string makes angle θ with the direction of motion of the block:

Rate of change of string length = $v \cdot \cos(\theta)$

This is used when string direction changes as block moves (non-ideal geometry).

SECTION 5: BLOCK ON INCLINE WITH PULLEY

Setup: Block (mass m) on incline of angle θ (smooth). String over massless pulley at top of incline.

String connects to hanging mass M .

Constraint: If M moves down by x , block moves up the incline by x (inextensible string).
 $\Rightarrow a_M$ (downward) = a_m (up the incline) = a

FBD and Newton's laws:

For M (downward positive): $Mg - T = Ma \dots (1)$

For m on incline (up incline positive): $T - mg\sin(\theta) = ma \dots (2)$ [smooth]

For m on rough incline: $T - mg\sin(\theta) - \mu mg\cos(\theta) = ma \dots (2')$

Adding (1) and (2) [smooth]: $(M - m\sin(\theta))g = (M + m)a$

$a = [M - m\sin(\theta)] \cdot g / (M + m)$

$T = M(g - a) = Mg(1 - (M - m\sin(\theta))/(M + m)) = Mm g(1 + \sin(\theta))/(M + m)$

Condition for no acceleration (equilibrium):

$a = 0$: $M = m\sin(\theta)$

Condition for block to move up (positive a for block):

$M > m\sin(\theta)$

Normal force on block: $N = mg\cos(\theta)$ [only if block moves along incline; no vertical accel component]

WORKED EXAMPLES — JEE ADVANCED LEVEL

Example 1: Atwood Machine — Complete Solution

Problem: $m_1 = 3$ kg, $m_2 = 5$ kg. $g = 10$ m/s². Find (a) acceleration, (b) tension, (c) velocity after 2 s.

Solution:

(a) $a = (m_2 - m_1)g / (m_1 + m_2) = (5 - 3) \cdot 10 / (3 + 5) = 20/8 = 2.5$ m/s²

(b) $T = 2m_1m_2g / (m_1 + m_2) = 2 \cdot 3 \cdot 5 \cdot 10 / 8 = 300/8 = 37.5$ N

(c) $v = u + at = 0 + 2.5 \cdot 2 = 5$ m/s (m_2 moving down, m_1 moving up)

Verification:

For m_2 : $5 \cdot 10 - 37.5 = 50 - 37.5 = 12.5$ N; $12.5 = m_2 \cdot a = 5 \cdot 2.5 = 12.5$. Correct.

For m_1 : $37.5 - 3 \cdot 10 = 37.5 - 30 = 7.5$ N; $7.5 = m_1 \cdot a = 3 \cdot 2.5 = 7.5$. Correct.

Example 2: Single Movable Pulley

Problem: Mass $M=4$ kg hangs from massless movable pulley. String passes over fixed pulley, one end fixed to ceiling. $g=10$.

Solution:

Two string segments support movable pulley. Let each have tension T .

Net upward force on (pulley + mass): $2T$. Weight: $Mg = 40$ N.

For massless pulley: $2T = Mg + Ma$ and $a_M = 2a_{\text{pulley}}$.

Treating the system: let $a_M = a$. $2T = M(g-a)$ [if mass moves down: net force = $Mg - 2T = Ma$].

String end is fixed: pulley is the free end. $a_{\text{string_end}} = 0 \Rightarrow a_M = 0$? No.

Actually string over fixed pulley, one end fixed: free end doesn't exist separately.

With one end fixed: $2T = Mg \Rightarrow T = Mg/2 = 20$ N. Mass in equilibrium if string can hold.

If string end is pulled: $T_{\text{applied}} = T$. $Mg - 2T = Ma$. With $MA=2$: $a_M = g - 2T/M$.

Example 3: Three-Mass Two-Pulley Constraint

Problem: Fixed pulley P_1 , movable pulley P_2 . Mass $m_1=6$ kg tied to axle of P_2 via string over P_1 .

Masses $m_2=2$ kg and $m_3=3$ kg hang on either side of string over P_2 .

Solution:

String 1 constraint: $a_{P_2} = -a_1$ (if m_1 goes up, P_2 goes down).

String 2 constraint: $a_2 + a_3 = 2a_{P_2} = -2a_1$.

Let T_1 = tension in string 1 (over P_1), T_2 = tension in string 2 (over P_2).

P_2 massless: $T_1 = 2T_2$.

For m_1 : $m_1g - T_1 = m_1a_1 \Rightarrow 60 - 2T_2 = 6a_1$... (A)

For m_2 : $m_2g - T_2 = m_2a_2 \Rightarrow 20 - T_2 = 2a_2$... (B)

For m_3 : $m_3g - T_2 = m_3a_3 \Rightarrow 30 - T_2 = 3a_3$... (C)

Constraint: $a_2 + a_3 = -2a_1$... (D).

From (B): $a_2 = (20 - T_2)/2$; from (C): $a_3 = (30 - T_2)/3$; sum: $(20 - T_2)/2 + (30 - T_2)/3 = -2(60 - 2T_2)/6$.

Solving: $3(20 - T_2) + 2(30 - T_2) = -2(60 - 2T_2) \Rightarrow 60 - 3T_2 + 60 - 2T_2 = -120 + 4T_2 \Rightarrow$

$120 - 5T_2 = -120 + 4T_2$.

$240 = 9T_2 \Rightarrow T_2 = 26.7$ N. $T_1 = 53.3$ N. $a_1 = (60 - 53.3)/6 = 1.1$ m/s².

Example 4: Block on Incline via Pulley

Problem: Block $m=2$ kg on smooth incline $\theta=30$ deg. Hanging mass $M=3$ kg. Find acceleration and tension.

Solution:

Constraint: $a_{\text{block(up incline)}} = a_M(\text{downward}) = a$.

For M (down+): $3g - T = 3a \Rightarrow 30 - T = 3a$... (1)

For m (up incline+): $T - 2g\sin 30 = 2a \Rightarrow T - 10 = 2a$... (2)

(1)+(2): $20 = 5a \Rightarrow a = 4$ m/s²

$T = 10 + 2 \cdot 4 = 18$ N

Check for m: $18 - 2 \cdot 10 \cdot 0.5 = 18 - 10 = 8$ N = $2 \cdot 4$. Correct.

Example 5: Massive Pulley (Rotational Inertia)

Problem: Atwood machine $m_1=2$ kg, $m_2=4$ kg, disk pulley $M=2$ kg, radius R . Find acceleration.

Solution:

Pulley has moment of inertia $I = (1/2)MR^2$ (for solid disk).

If pulley rotates: net torque $= (T_2 - T_1)R = I\alpha = I(a/R) = (1/2)MR^2(a/R) = (1/2)MRa$.

$T_2 - T_1 = (1/2)Ma$... (torque equation)

For m_2 : $m_2g - T_2 = m_2a$... (ii)

For m_1 : $T_1 - m_1g = m_1a$... (iii)

(ii)+(iii)+(torque): $(m_2 - m_1)g = (m_1 + m_2 + M/2)a$

$a = (4 - 2) \cdot 10 / (2 + 4 + 1) = 20/7 = 2.86 \text{ m/s}^2$

$T_1 = m_1(g + a) = 2 \cdot (10 + 20/7) = 2 \cdot 90/7 = 25.7 \text{ N}$

$T_2 = m_2(g - a) = 4 \cdot (10 - 20/7) = 4 \cdot 50/7 = 28.6 \text{ N}$

Example 6: Complex Pulley — Acceleration Ratio

Problem: Block A (3 kg) on table, block B (5 kg) and block C (2 kg) hang from two strings over

two pulleys. String from A goes left over pulley to B. String from A goes right over pulley to C.

Table is smooth. Find accelerations.

Solution:

Constraints: a_A (left) = a_B (down) = a ; a_A (right) = a_C (down) = b .

Wait — two separate strings, so a_A overall: must satisfy both simultaneously.

If table is smooth: only one degree of freedom per string.

Actually with two strings over two separate pulleys, these are independent systems.

String 1 (A-B): A moves right, B moves down. Constraint: $a_B = a_{A_right}$.

String 2 (A-C): A moves left, C moves down. Constraint: $a_C = a_{A_left}$.

But A can only have one acceleration! This means one string goes taut, other slack, or $a=b$.

For equilibrium: $T_B - T_C = m_A \cdot a_A$. $m_B g - T_B = m_B a$; $m_C g - T_C = m_C a$.

$5 \cdot 10 - T_B = 5a$; $T_B - T_C = 3a$; $2 \cdot 10 - T_C = 2a$.

Adding all: $(5+2) \cdot 10 = (5+3+2)a \Rightarrow 70 = 10a \Rightarrow a = 7 \text{ m/s}^2$.

$a_A = a = 7 \text{ m/s}^2$, $a_B = 7 \text{ m/s}^2$ downward, $a_C = 7 \text{ m/s}^2$ downward.

PRACTICE QUESTIONS — JEE ADVANCED LEVEL

Q1. In an Atwood machine $m_1=1$ kg, $m_2=3$ kg. Acceleration of the system ($g=10$) is:

(A) 5 m/s^2 (B) 2.5 m/s^2 (C) 10 m/s^2 (D) $g/4$

Q2. For Atwood machine with $m_1=m_2=m$, the tension in the string is:

(A) mg (B) $mg/2$ (C) $2mg$ (D) 0

Q3. In a movable pulley system, if the free end of the string is pulled with velocity v , the load rises with velocity:

(A) v (B) $v/2$ (C) $2v$ (D) $v/4$

Q4. Block A (3 kg) on smooth table connected by string over pulley to hanging block B (2 kg). Acceleration is:

(A) 4 m/s^2 (B) 2 m/s^2 (C) $g/3$ (D) $g/4$

Q5. A movable pulley with string tension $T=20$ N supports a load. The load weight is:

(A) 10 N (B) 20 N (C) 40 N (D) 30 N

Q6. Atwood machine $m_1=2$ kg, $m_2=3$ kg. Ratio of tension to weight of lighter mass $T/(m_1g)$ is:

(A) $6/5$ (B) $5/6$ (C) $12/5$ (D) $5/12$

Q7. For block m on smooth incline (30 deg) connected to hanging mass M via pulley ($a=0$), M/m equals:

(A) 1 (B) $1/2$ (C) $\sin 30 = 0.5$ (D) $\cos 30$

Q8. If a pulley system has 4 movable pulleys in cascade, the mechanical advantage is:

(A) 4 (B) 8 (C) 16 (D) 2

Q9. When $m_1=m_2$ in Atwood machine and a small mass dm is added to m_2 , initially at rest. Acceleration starts from:

(A) Zero and increases (B) g immediately (C) $dm \cdot g / (2m + dm)$ (small) (D) $g/2$

Q10. Two blocks $m_1=2$ kg and $m_2=3$ kg connected by string over frictionless pulley, m_1 on smooth incline 45 deg. Acceleration:

(A) $(3-2\sin 45) \cdot 10/5$ (B) $(3+2\sin 45) \cdot 10/5$ (C) $g/2$ (D) g

ANSWER KEY AND EXPLANATIONS

Q1 — Answer: (A) 5 m/s^2

$$a = (3-1) \cdot 10 / (1+3) = 20/4 = 5 \text{ m/s}^2.$$

Q2 — Answer: (A) mg

$$T = 2 \cdot m \cdot g / (m+m) = 2m^2 \cdot g / (2m) = mg. \text{ System is in equilibrium; no acceleration.}$$

Q3 — Answer: (B) $v/2$

$$\text{Movable pulley: load speed} = \text{half string-end speed. } v_{\text{load}} = v/2.$$

Q4 — Answer: (A) 4 m/s^2

$$\text{Net force} = m_B \cdot g = 20 \text{ N. Total mass} = 3+2 = 5 \text{ kg. } a = 20/5 = 4 \text{ m/s}^2.$$

Q5 — Answer: (C) 40 N

$$\text{Two segments: upward force} = 2T = 40 \text{ N} = \text{load weight.}$$

Q6 — Answer: (A) $6/5$

$$T = 2 \cdot 2 \cdot 3 \cdot 10 / (2+3) = 120/5 = 24 \text{ N. } m_1 \cdot g = 20 \text{ N. Ratio} = 24/20 = 6/5.$$

Q7 — Answer: (C) $M = m \cdot \sin 30 = m/2$

$$\text{Equilibrium: } T = Mg, T = m \cdot g \cdot \sin(30) = mg/2. \text{ So } M = m/2.$$

Q8 — Answer: (C) 16

$$n \text{ movable pulleys: } MA = 2^n = 2^4 = 16.$$

Q9 — Answer: (C) $dm \cdot g / (2m + dm)$

$$a = (m_2 + dm - m_1) \cdot g / (m_1 + m_2 + dm) = (dm) \cdot g / (2m + dm). \text{ Small initially.}$$

Q10 — Answer: (A) $(3 - 2 \cdot \sin 45) \cdot 10/5$

$$\text{Net force} = m_2 \cdot g - m_1 \cdot g \cdot \sin 45 = (3 - 2 \cdot 0.707) \cdot 10 = (3 - 1.414) \cdot 10 = 15.86 \text{ N. Total mass} = 5 \text{ kg. } a = 3.17 \text{ m/s}^2.$$